■ SOLAR SYSTEM SCALE MODEL

A Space Scale Safari — Detailed Lesson Plan

Subject	Science — Earth & Space
Grade / Class	Grade 4 (Age 9–10)
Duration	90 Minutes (2 periods)
Author	Niranjanadevi M
Curriculum Link	ISRO Missions · Scale & Measurement · Solar System

"If you could drive a car to the Sun, it would take over 170 years! Today we shrink the Solar System to fit in our classroom."

8 Planets | **Scale Models** | **Hands-On** | **ISRO Focus**

LESSON OVERVIEW

Topic	Solar System Scale Model: Sizes and Distances of Planets
Subject	Environmental Science / Science — Earth & Space
Grade	Grade 4 (Age 9–10 years)
Duration	90 minutes (can be split across 2 × 45-min periods)
Class Size	20–40 students (adaptable)
Setting	Classroom + School ground / corridor (for distance activity)
Prior Knowledge	Names of planets, basic measurement (metres, centimetres)

LEARNING OBJECTIVES

K

- Name all 8 planets in order from the Sun
- State the scale used in the model (1 AU = 1 metre)
- Recall at least 3 facts about ISRO space missions

U

- Explain why scientists use scale models
- Describe how planet sizes and distances compare
- Relate Indian everyday objects to planet sizes

S

- Measure and mark distances using a tape measure / string
- Create a scale model using classroom / household materials
- Record observations in a structured worksheet

V

- Appreciate the vastness and wonder of the universe
- Feel pride in India's achievements through ISRO
- Develop curiosity and scientific thinking

CURRICULUM LINKS & 21st CENTURY SKILLS

NCERT EVS / Science Ch.	Our Solar System · Stars and the Solar System (Class 4–5 EVS)
Mathematics	Measurement, ratio, scale, basic multiplication
21st-C Skills	Critical Thinking · Collaboration · Creativity · Communication (4Cs)
NEP 2020 Alignment	Experiential learning, hands-on inquiry, STEM integration
SDG Link	SDG 4 — Quality Education · SDG 17 — Science Literacy

■ MATERIALS & PREPARATION

Teacher Preparation Checklist (tick when ready)

- ☐ Measuring tape (30 m) or long string
- ☐ Mustard seeds (Mercury)
- ☐ Laddoo / foam ball 5 cm (Jupiter)
- ☐ Large kadhai / pot (Sun reference)
- ☐ Printed student worksheets
- ☐ Slideshow / printed slides (provided)
- ☐ Stickers for reward (optional)
- ☐ String 35 m (for ground model)
- ☐ Chalk or marker sticks (8 pcs)
- ☐ Peppercorns × 8 (Earth)
- ☐ Cricket ball (Saturn)
- ☐ Planet name cards (print below)
- ☐ Whiteboard / display board
- ☐ Quiz answer cards
- ☐ Timer or stopwatch
- ☐ Colour pencils for worksheet

Room Setup: Arrange desks in groups of 4–5 for collaborative work. Keep a 2 m clear space at the front for demonstrations. For the outdoor distance activity, identify a straight 35-metre path on the school ground.

■ PLANET REFERENCE DATA TABLE

Planet	Order	Real Distance from Sun	Scale Distance (1 AU = 1 m)	Scale Size (Indian Object)	Fun Colour Code
Mercury ■	1st	57.9 M km	0.4 m	Mustard seed	
Venus ■	2nd	108.2 M km	0.72 m	Lentil (dal)	
Earth ■	3rd	149.6 M km	1.00 m	Peppercorn	
Mars ■	4th	227.9 M km	1.52 m	Small marble	
Jupiter ■	5th	778.6 M km	5.20 m	Laddoo (5 cm)	
Saturn ■	6th	1,427 M km	9.58 m	Cricket ball	
Uranus ■	7th	2,871 M km	19.2 m	Tennis ball	
Neptune ■	8th	4,498 M km	30.1 m	Baseball	

Solar System distance scale — each planet's position relative to the Sun

LESSON PROCEDURE — STEP-BY-STEP TIMELINE

1 HOOK — Welcome to Space!

0–10 min

- Show slide 1 (cover) as students enter — play soft space music if possible
- Ask: 'Has anyone seen the Moon at night? How far do you think it is?'
- Reveal ISRO fact: 'India sent Chandrayaan to the Moon and Mangalyaan to Mars!'
- Pose the big question: 'If the Sun were 1 metre, how far away is Neptune?'
- Set excitement: 'Today we shrink the Solar System to fit in our school!'

2 CONCEPT — Why Scale Models?

10–20 min

- Show slides: 'Why Use Scale Models?' and 'Scientists use models...'
- Explain: The Solar System is so huge we can't see it all at once
- Introduce the scale: 1 AU (Sun to Earth) = 1 metre in our model
- Quick check: 'What is 1 AU? How far is that in real life?' (149.6 million km)
- Display the Planet Reference Table — let students read it with partners

3 DIRECT TEACHING — Meet the 8 Planets

20–30 min

- Teach the mnemonic: My Teacher (like a mother) Teaches Everyone Properly
- Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune
- Hold up each Indian object: mustard seed, peppercorn, laddoo, cricket ball
- Compare sizes dramatically: 'Over 1,000 Earths fit inside Jupiter!'
- Show Sun comparison: 'Our big kadhai = Sun; peppercorn = Earth'
- Ask students to draw the 8 planets in order in their notebooks

4 HANDS-ON — School Ground Distance Walk

30–50 min

- Move to the school ground with a 30 m tape and 8 planet marker cards
- Designate one student as 'The Sun' — they stand at the start
- Call up students one by one to walk to their planet's distance:
- Mercury: 0.4 m · Venus: 0.72 m · Earth: 1 m · Mars: 1.52 m
- Jupiter: 5.2 m · Saturn: 9.58 m · Uranus: 19.2 m · Neptune: 30 m
- Each student plants a chalk mark and holds their planet card
- Ask: 'Feel the space between Jupiter and Saturn! What do you notice?'
- Photograph the class for display — return inside for reflection

5 ACTIVITY — Classroom Scale Model

50–65 min

- Distribute worksheet and household objects (mustard seed, peppercorn, etc.)
- Students arrange objects on their desk in order, spaced proportionally
- Complete Part A of worksheet: draw and label each planet with its object
- Groups compare their models — discuss differences
- Teacher circulates, asks probing questions (see Discussion section)

6

ENRICHMENT — ISRO Inspires Us!

65–75 min

- Show ISRO missions slide: Chandrayaan-1 (2008), Mangalyaan (2013),
- Chandrayaan-3 (2023 — South Pole landing!), Aditya-L1 (Sun study)
- Calculate together: Mangalyaan travelled Earth→Mars = 670 million km
- In our model that is just 1.12 m (Mars distance minus Earth distance)
- Ask: 'What would YOU want to explore in space?'
- Show the inspiration slide: 'You too can be a space explorer!'

7

ASSESSMENT — Pop Quiz + Worksheet Part B

75–85 min

- Oral quick-fire quiz — raise hands (see quiz questions below)
- Students complete worksheet Part B (fill in the blanks, match column)
- Peer-check answers after 5 minutes
- Award stickers for correct answers (optional)

8

CLOSURE — Recap & Homework

85–90 min

- Recap key facts: order, scale (1 AU = 1 m), space is mostly empty
- Show the 'Quick Recap' slide — students recite facts chorally
- Homework: Create your own scale model at home with family
- Use household items; share a photo or drawing next class
- Exit ticket: write ONE thing you learned and ONE question you have

DISCUSSION QUESTIONS

Opening / Recall

Q: What are the 8 planets of our Solar System? Can you name them in order?

Q: Which planet is closest to the Sun? Which is farthest?

Q: What is a scale model? Can you think of a scale model you have seen before?

During the Activity

Q: Why does the space between the outer planets feel so much bigger?

Q: If a peppercorn is Earth, how big do you think the Sun really is compared to us?

Q: Why do scientists use AU (Astronomical Units) instead of kilometres?

Q: If you could travel at car speed (100 km/h), how long would it take to reach Mars?

Higher-Order Thinking

Q: If we added a 9th planet to our model, where do you think it would be?

Q: Why is it important to have scale models — what would happen if we couldn't?

Q: ISRO's Mangalyaan orbited Mars. On our model, how far did it travel from Earth?

Q: If Earth is a peppercorn, what everyday object could represent the entire Milky Way?

ISRO Connections

Q: Which ISRO mission landed on the Moon in 2023?

Q: Mangalyaan reached Mars in 298 days. How did scientists plan that journey?

Q: What is Aditya-L1 studying, and why is studying the Sun important?

Q: How does India's space programme inspire young students like you?

DIFFERENTIATION & INCLUSIVE STRATEGIES

Support (Below Level)	Core (On Level)	Extension (Above Level)	EAL / ESL Learners
• Pre-labelled planet cards • Partner support • Simplified worksheet • Reduced planet count (4) • Verbal answers accepted	• Full 8-planet model • Standard worksheet • Group discussion • Mnemonic practice • Quiz participation	• Research additional facts • Calculate travel times • Design a 9th planet • Write ISRO mission log • Present to the class	• Bilingual planet cards • Visual/icon worksheet • Tamil/Hindi labels • Buddy pairing • Extra time allowed

■ ASSESSMENT & EVALUATION

Assessment Type	When	Tool / Method	What to Look For
Diagnostic	Start of lesson	Oral Q&A;	Prior knowledge of planet names, basic space facts
Formative	During activity	Observation checklist	Correct ordering, accurate distances, teamwork
Formative	During walk	Teacher observation	Understanding of scale, spatial reasoning
Summative	End of lesson	Worksheet + Quiz	Recall of 8 planets, scale concept, ISRO facts
Self-Assessment	Exit ticket	Written reflection	'One thing I learned / One question I have'
Portfolio	Take home	Photograph of model	Creativity, accuracy, family engagement

■ POP QUIZ — QUESTIONS & ANSWERS

Q1. Which planet is closest to the Sun?

A: Mercury — only 0.4 m in our model (57.9 million km in reality)

Q2. Which planet is farthest from the Sun?

A: Neptune — 30 m in our model (4,498 million km in reality)

Q3. What object represents Earth in our Indian scale model?

A: A Peppercorn — Earth is only 12,742 km in diameter

Q4. How far is Jupiter from the Sun in our school ground model?

A: 5.2 metres (778.6 million km in reality — over 5 times Earth's distance)

Q5. What does '1 AU' mean?

A: 1 Astronomical Unit = distance from Sun to Earth = 149.6 million km

Q6. Name ONE ISRO mission to the Moon.

A: Chandrayaan-1 (2008) OR Chandrayaan-3 (2023, landed near south pole)

Q7. Which ISRO mission went to Mars?

A: Mangalyaan (Mars Orbiter Mission, 2013) — India's first interplanetary mission

Q8. Why do scientists use scale models?

A: The Solar System is too big to see all at once; models help us understand sizes and distances

■■ ISRO ENRICHMENT — INDIA IN SPACE

■ **Chandrayaan-1**	**2008**	India's first lunar mission — discovered water ice on the Moon
■ **Mangalyaan**	**2013**	Mars Orbiter Mission — India 1st nation to succeed on first attempt
■ **Chandrayaan-3**	**2023**	Landed near Moon's South Pole — India 4th country to soft-land on Moon
■■ **Aditya-L1**	**2024**	Studying the Sun from Lagrange Point 1 — India's first solar mission

"Reach for the stars, like ISRO!"

India's space agency has shown us that with hard work and dreams, we can achieve anything amazing. From launching rockets to reaching the Moon and Mars, ISRO proves that Indian scientists are among the best in the world. **You too can be a space explorer one day!**

STUDENT WORKSHEET — Printable

Name: ________________ **Class / Section:** ___________ **Date:** _______________ **Score:** ___ / 20

PART A — Fill in the Scale Distance (8 marks)

Write the correct scale distance (in metres) for each planet in our model.

Planet	Order from Sun	Your Answer (metres)	Correct Answer
Mercury	1st	___________ m	0.4 m
Venus	2nd	___________ m	0.72 m
Earth	3rd	___________ m	1.00 m
Mars	4th	___________ m	1.52 m
Jupiter	5th	___________ m	5.20 m
Saturn	6th	___________ m	9.58 m
Uranus	7th	___________ m	19.2 m
Neptune	8th	___________ m	30.1 m

PART B — Match the Indian Object to the Planet (6 marks)

Planet	Your Match (Object)	Column B — Objects
Mercury	_______	A. Laddoo (5 cm)
Earth	_______	B. Cricket Ball
Jupiter	_______	C. Mustard Seed
Saturn	_______	D. Peppercorn
Sun	_______	E. Big Kadhai
Mars	_______	F. Small Marble

ANSWER KEY (Match): Mercury → C · Earth → D · Jupiter → A · Saturn → B · Sun → E · Mars → F

PART C — Short Answer (6 marks)

1. Why do scientists use scale models?

Answer: ____

Model Answer: The Solar System is too large to observe all at once; models shrink it to a manageable size.

2. Name TWO ISRO missions and where they went.

Answer: ____

Model Answer: Any two from: Chandrayaan-1/3 (Moon), Mangalyaan (Mars), Aditya-L1 (Sun/L1 point).

3. If the Sun is 1 metre in our model, how big is Jupiter?

Answer: ____

Model Answer: Jupiter would be about 10 cm across (1/10th of 1 metre).

■ TEACHER NOTES & TIPS

Common Misconceptions to Address

- Students often think planets are equally spaced — the outdoor walk powerfully corrects this.
- Many think the Sun is 'slightly bigger' than planets — emphasise the kadhai vs peppercorn scale.
- Students may confuse 'AU' with 'a unit' — clarify it specifically means Sun-to-Earth distance.
- Outer planet distances feel abstract — have students actually walk 30 m for Neptune.

Classroom Management Tips

- Assign roles for the ground activity: Measurer, Marker, Card-holder, Photographer, Timekeeper.
- Use a buddy system — each planet student pairs with another for the return walk.
- Keep the energy up during the walk — narrate each planet like a space mission commentary!
- If ground space is limited, scale down: 1 AU = 50 cm (Neptune at 15 m).

Extension Activities

- Research project: 'Design your own space mission — where would you go and why?'
- Art: Paint the Solar System to scale on a long roll of paper.
- Math link: Calculate how long it would take to drive to each planet at 100 km/h.
- Writing: Write a journal entry as an astronaut on Chandrayaan-3 approaching the Moon.
- Science fair: Build a 3D scale model solar system for the school corridor.

Safety & Logistics

- Ensure the outdoor area is safe, clear of obstacles, and supervised by additional staff.
- Bring sun protection (hats/sunscreen) if activity is in direct sunlight.
- Keep students together — use a buddy system and establish a clear signal to return.
- Have a backup plan for rainy days: use the school corridor or mark distances on the classroom floor.

■ REFERENCES & FURTHER READING

[1] ISRO Official Website — isro.gov.in — Mission details for Chandrayaan and Mangalyaan

[2] NASA Solar System Exploration — solarsystem.nasa.gov — Planet data and scale model guides

[3] NCERT EVS Class 4 Chapter: 'Our Solar System' — textbook pages on planet order

[4] Jet Propulsion Laboratory (JPL) — jpl.nasa.gov/edu — Classroom scale model resources

[5] National Science Foundation — 'If the Sun Were a Grapefruit' classroom activity

[6] Niranjanadevi M — Space Scale Safari presentation (Canva, 2026) — source material

Solar System Scale Model — Space Scale Safari

Lesson Plan prepared by Niranjanadevi M • Grade 4 Science • 2026

"Keep looking up and dreaming big! Who knows — maybe one day you'll be an astronaut or scientist at ISRO! ■"